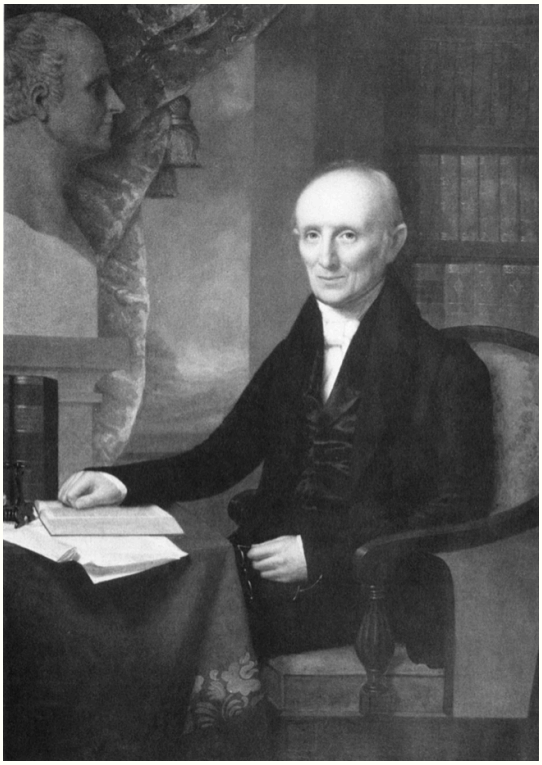
Nathaniel Bowditch: the self-taught navigator

Faria lesson · 2026-11-06 · faria.jbf.com/topics/49-bowditch/

Jeff asked who Bowditch was after we talked about Polaris correction tables. This page is the full answer. It pairs with Polaris (the star whose tiny offset Bowditch's tables correct), The sidereal day and latitude sailing (why the correction is needed, and Jeff's "oscilloscope wave" insight), Galileo (longitude as time, before chronometers), and Map Surprises.



Portrait by Charles Osgood. Public domain (Peabody Essex Museum), via Wikimedia Commons.

1. Salem childhood: school ends at ten

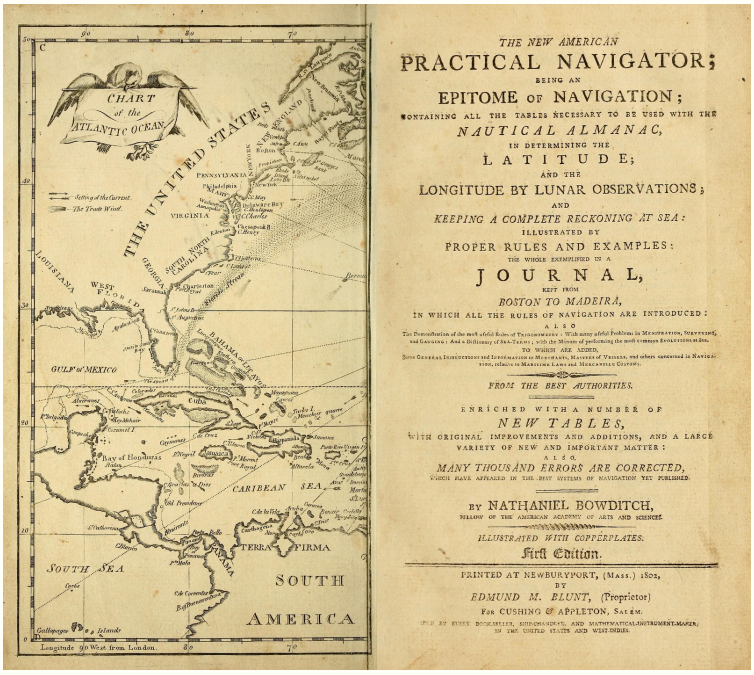
Nathaniel Bowditch was born in Salem, Massachusetts, on 26 March 1773, the fourth of seven children of Habakkuk Bowditch, a cooper (and sometime sailor who quit the sea after his ship went aground in 1775), and Mary Ingersoll Bowditch.

Family money ran short. At about age ten his formal schooling stopped and he went to work in his father's cooperage. At twelve he was indentured for nine years as a bookkeeping apprentice to a ship chandler — a dealer in rope, canvas, tar, and the other stores a ship needs. He lived and worked in the chandlery and learned accounts.

That "left school at ten" is the fact Jeff asked about. It is confirmed in the standard biographies (MacTutor / Biographical Encyclopedia of Astronomers; Wikipedia following Henry Ingersoll Bowditch's memoir of his father).

2. Self-taught: algebra, calculus, Latin, French, Newton's Principia

In the chandlery Bowditch taught himself.



1802 first edition. Public domain (U.S. Naval Observatory), via Wikimedia Commons.

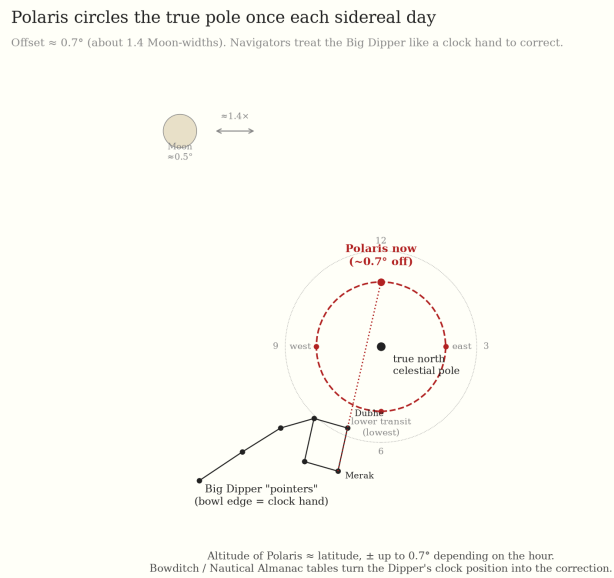
- 1786 (age 14): algebra.
- 1788: calculus.
- 1790: Latin, specifically so he could read *Isaac Newton's Philosophiae Naturalis Principia Mathematica.
- 1792: French, for the European mathematical literature.

At eighteen he copied out all the mathematical papers in the Philosophical Transactions of the Royal Society. In 1793 he found an error in Newton's Principia.

Serendipity helped. In 1781 the Beverly privateer Pilgrim captured a ship carrying the scientific library of the Irish chemist Richard Kirwan — Newton, the Bernoullis, Chambers's Cyclopedia, and more. The books were bought for Salem and eventually housed with the Rev. John Prince, who in 1791 let the eighteen-year-old Bowditch use what had become the best scientific library north of Philadelphia. Without that captured library, the self-education would have been far harder.

3. Five voyages, 1795–1803

Bowditch went to sea:



Polaris's ~0.7° circle and the Big Dipper clock. Drawn for faria.jbf.com.

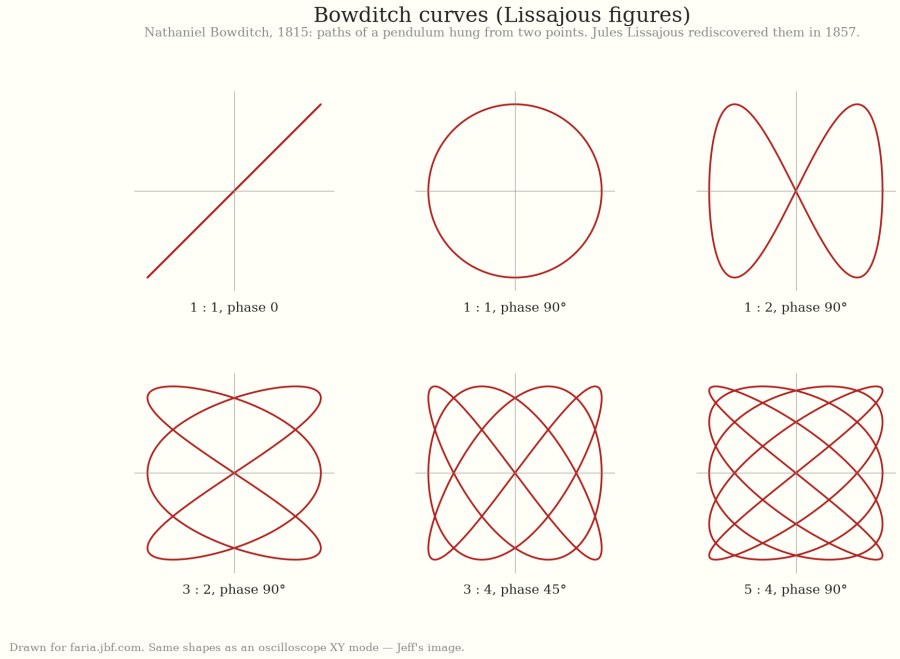
- Voyage — Role — Notes
- 1st — Ship's clerk / second mate — Began finding errors in Moore's tables; conceived a clearer method of lunars
- 2nd–4th — Supercargo — Continued corrections; taught crew navigation
- 5th — Master and part owner — Last voyage; returned to Salem 1803

Between the first and second voyages, in March 1798, he married Elizabeth Boardman; she died seven months later while he was at sea. In 1800 he married his cousin Mary "Polly" Ingersoll; they had eight children.

After 1803 he left the sea for mathematics, insurance, and the book that made his name.

4. ~8,000 errors in Moore's Practical Navigator

The standard English handbook was John Hamilton Moore's The Practical Navigator (London). Its tables were known to be faulty. On his first voyage Bowditch began checking them. By the time he was done he had compiled a list of more than 8,000 errors.



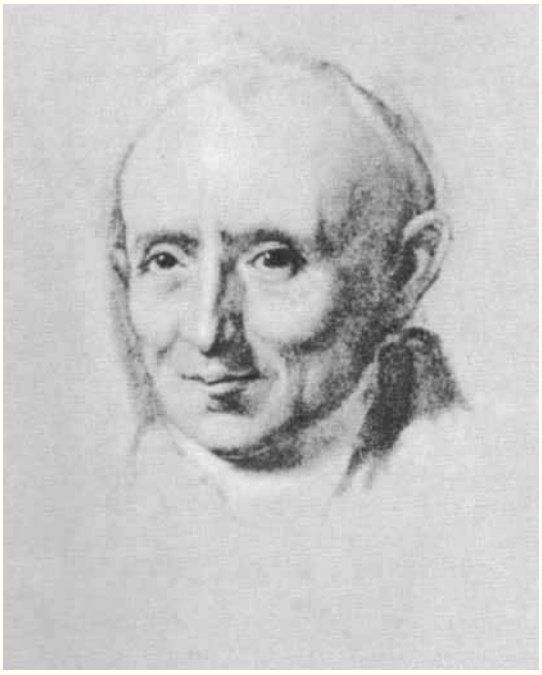
Bowditch curves (Lissajous figures), 1815 — oscilloscope XY shapes. Drawn for faria.jbf.com.

A careful caveat (from a modern study of Moore and Bowditch in the Journal of Navigation): most of those errors were small — last-decimal-place mistakes in traverse tables, proportional logarithms, and log-trig functions. Saying "8,000 errors" can make Moore sound like a fraud; he was not. But at sea a wrong last digit can still put you on a reef. Bowditch recomputed everything.

He first sent corrections to the American publisher Edmund March Blunt. Blunt issued corrected American editions of Moore in 1799 and 1800. By the third round the book had changed so much that Blunt published it under Bowditch's own name.

5. The New American Practical Navigator (1802)

In 1802 Blunt published Nathaniel Bowditch, The New American Practical Navigator, printed at Newburyport, Massachusetts. Bowditch's rule for what went in:



Unfinished Gilbert Stuart portrait, 1828. Public domain, via Wikimedia Commons.

> *"put down in the book nothing I can't teach the crew."*

The book rearranged Moore, added tables, and included Bowditch's clearer solutions for the spherical triangle and for lunar distances — finding longitude by measuring the Moon's angular distance from the Sun or a star, then comparing with an almanac. Accurate marine chronometers (Harrison's) existed but were still too expensive for most merchant captains; lunars were the poor man's clock in the sky. (See Galileo for the older dream of a Jupiter-moon clock.)

Still in print. In 1867–68 the U.S. Hydrographic Office bought the copyright. The book has been in continuous government publication ever since. Today it is NGA Pub. No. 9, The American Practical Navigator, issued free by the National Geospatial-Intelligence Agency in two volumes (2024 edition as of this writing). Every commissioned U.S. Navy ship still carries it.

Sailors simply call it "Bowditch" — "the bible of the sea." Midshipmen before the Naval Academy existed called it "the immaculate Bowditch."

6. Teaching the crew — including the cook

Bowditch's teaching story is the one Jeff heard.

On the fourth voyage, to Manila in the ship Astræa, Captain Henry Prince was asked in Manila how an American ship had found its way so well against the northeast monsoon. Prince answered (as recorded in Henry Ingersoll Bowditch's memoir of his father) that he had a crew of twelve men, every one of whom could

> *"take and work a lunar observation as well, for all practical purposes, as Sir Isaac Newton himself were he alive."*

Bowditch sat through the conversation "as modest as a maid," slate pencil in his mouth. Another merchant added that there was "more knowledge of navigation on board that American ship than there has been in all the ships that ever came into Manila Bay."

Later tellings (including Baron Franz von Zach and many navy histories) add that even the ship's cook became proficient in lunar distances. The cook detail is traditional and widely repeated; the core claim — an entire twelve-man crew able to work lunars — is anchored in Captain Prince's contemporary boast. Bowditch's own standard for the book was the same spirit: write nothing you cannot teach the forecastle.

7. How Bowditch corrects Polaris

This is the bridge to Polaris and to Jeff's insight on the sidereal day page.

Polaris is not exactly on the north celestial pole. It sits about 0.7° off (roughly 1.4 Moon-widths), and therefore circles the true pole once each sidereal day. If you take the raw altitude of Polaris as your latitude, you can be wrong by up to 0.7°, about 42 nautical miles (since 1° of latitude = 60 nmi).

Navigators fix that with a Polaris correction. The classic rule of thumb treats the Big Dipper's pointer stars (Dubhe and Merak, the outer edge of the bowl) like a clock hand around the pole. From the "hour" on that clock you look up a correction in the tables — in Bowditch historically, and today in the Nautical Almanac (corrections a0, a1, a2) — and add or subtract it. The wave flattens. The altitude of Polaris, corrected, is your latitude.

8. Laplace's Mécanique céleste

Bowditch taught himself French partly to read Pierre-Simon Laplace's monumental Traité de mécanique céleste. He finished translating the first four volumes by about 1818, then spent years adding commentaries that roughly doubled the length: supplying steps Laplace had skipped, inserting later results, and giving credits Laplace had omitted. Publication (at Bowditch's own expense) ran 1829–1839. The translation became a foundation of American physical astronomy in the nineteenth century. Benjamin Peirce helped with the later volumes.

Harvard offered him the chair of mathematics and physics in 1806; West Point and the University of Virginia also offered chairs. He turned them all down — the insurance salary was higher, and he preferred the work.

9. Actuary and "money manager"

In 1804 Bowditch became president of the Essex Fire and Marine Insurance Company in Salem — often called America's first insurance actuary in the modern sense. The company prospered through the War of 1812. In 1823 he moved to Boston as actuary for the Massachusetts Hospital Life Insurance Company, investing the fortunes of sea-made wealth into manufacturing towns such as Lowell. When he moved, he took more than 2,500 books, 100 maps, and 29 volumes of his own manuscripts.

He was elected to the American Academy of Arts and Sciences (1799), the American Philosophical Society (1809), and later the Royal Societies of London and Edinburgh and the Royal Irish Academy. In 1829 he became the first American foreign associate of the Royal Astronomical Society. Harvard gave him an honorary degree in 1802 and an LL.D. in 1816.

10. Bowditch curves = Lissajous figures = Jeff's oscilloscope

In 1815 Bowditch published "On the Motion of a Pendulum Suspended from Two Points" in the Memoirs of the American Academy of Arts and Sciences. A pendulum hung from two points (a Y-shaped or Blackburn suspension) can swing at different frequencies in two perpendicular directions. The bob traces the curves

$$x = a\,\sin(n\,t + c), \quad y = b\,\sin(t)$$

Jules Antoine Lissajous studied them independently and more famously in 1857, using light reflected from tuning forks; his name stuck. Because Bowditch was first, mathematicians also call them Bowditch curves.

Jeff's connection. When Jeff pictured an uncorrected Polaris track as "cycles on an oscilloscope," he reached for exactly this family of shapes. An oscilloscope in XY mode draws Lissajous / Bowditch figures. The navigator who wrote the bible of the sea also drew the curves on the screen Jeff was imagining. See The sidereal day and latitude sailing for the wavy track itself.

11. Death, eulogy, and legacy

Bowditch died in Boston on 16 March 1838 of stomach cancer. He is buried in Mount Auburn Cemetery, Cambridge; the memorial statue by Robert Ball Hughes was the first life-size bronze cast in America.

The Salem Marine Society eulogized him:

> *"In his death a public, a national, a human benefactor has departed. Not this community nor our country only, but the whole world has reason to do honor to his memory. When the voice of eulogy shall cease to flow, no monument will be needed to keep alive his memory among men; but as long as ships shall sail, the needle point to the north, and the stars go through their wonted courses in the heavens, the name of Dr. Bowditch will be revered as of one who has helped his fellowmen in time of need, who was and is a guide to them over the pathless oceans, and one who forwarded the great interests of mankind."*

Named for him: the oceanographic survey ship USNS Bowditch; a lunar crater; a Salem–Boston ferry; a Maine windjammer; schools; the Department of Marine Transportation building at the U.S. Merchant Marine Academy; and Jean Lee Latham's 1955 Newbery Medal novel Carry On, Mr. Bowditch.

Review questions

1. Why did Bowditch teach himself Latin?
2. What was his rule for what belonged in the Navigator?
3. Why does Polaris need a correction table, and what "clock" do sailors use?
4. What do Bowditch curves have to do with an oscilloscope?

Key facts

- Nathaniel Bowditch, Salem, 26 Mar 1773 – Boston, 16 Mar 1838.
- Left school ~age 10; indentured at 12 to a ship chandler for nine years; self-taught algebra, calculus, Latin (for Newton), French (for Laplace).
- Five voyages 1795–1803; found more than 8,000 errors in J. H. Moore's Practical Navigator (mostly small tabular mistakes).
- The New American Practical Navigator, 1802; still published as NGA Pub. No. 9; sailors call it "Bowditch."
- Rule: "put down in the book nothing I can't teach the crew"; on the Manila voyage the whole twelve-man crew (cook included, by tradition) could work lunars.
- Polaris correction: star is ~0.7° off the pole; Big Dipper pointers as clock; tables remove up to ~42 nmi of latitude error.
- Translated and annotated Laplace's Mécanique céleste (vols. I–IV, pub. 1829–39); declined Harvard / West Point / Virginia chairs.
- America's early insurance actuary (Essex Fire & Marine, then Mass. Hospital Life).
- Bowditch curves (1815) = Lissajous figures (1857) = oscilloscope XY shapes.
- USNS Bowditch; Mount Auburn bronze; Salem Marine Society eulogy: "as long as ships shall sail…"

Answers

1. So he could read Newton's Principia in the original Latin.
2. "Put down in the book nothing I can't teach the crew."
3. Polaris is ~0.7° from the true pole and circles it once per sidereal day; raw altitude can be wrong by up to ~42 nmi. Sailors use the Big Dipper pointers as a clock hand and look up the correction in Bowditch / the Almanac.
4. They are the same family of curves: Bowditch studied them in 1815 as pendulum paths; an oscilloscope in XY mode draws them live.